LRS^2-DM: Small Ship Target Detection in Low-Resolution Remote Sensing Images Based on Diffusion Models

Yantong Chen, Jingyu Yan, Yifan Liu, and Zhi Gao

***Abstract*— With advancements in remote sensing technology, ship detection has emerged as a pivotal component in marine environmental protection and maritime traffic management. However, the significant distance of satellite imaging results in ship targets appearing as small-scale objects in the images. Current detection algorithms face challenges in accurately identifying the features of small ship targets in low-resolution settings. Therefore, this article proposes a small ship target detection model for low-resolution remote sensing images based on diffusion models (DMs). In the first stage, cognitive conditions are used as inputs. A low-level super-resolution (L^{2}SR) module enhances image clarity and facilitates the extraction of richer ship target features. The second stage employs a spatial refinement module (SRM) to effectively enhance textures, edges, and other fine-grained features of small targets. Finally, an optimized loss function is designed to mitigate uncertainties arising from noise in the DM for remote sensing images. The experimental results demonstrate that the proposed method achieves superior performance on the DOTA-v2.0-Ship and S-Ship datasets, attaining average precision (AP) values of 95.34% and 96.12%, respectively. Moreover, it sustains a high frame-per-second (FPS) rate, striking an optimal balance between detection accuracy and computational efficiency.**

***Index Terms*— Diffusion models (DMs), low resolution, ship inspection, small goals.**

I. Introduction

THE rapid development of remote sensing technology has significantly improved daily life convenience and enhanced safety measures. Leveraging its robust capability for long-distance target detection and identification, remote sensing technology plays a pivotal role in ocean monitoring. Remote sensing technology, when applied to ship detection, aids law enforcement in combating maritime crimes. It helps maintain oceanic order and is essential for ensuring maritime security.

However, due to the long imaging distance of remote sensing satellites [1] and the impact of atmospheric disturbances during the imaging process, low-resolution images are often produced. These limitations pose significant challenges to target detection. Remote sensing ship targets are represented by a limited number of pixels. Their small size further increases the difficulty of target detection. This challenge is demonstrated in Fig. 1. Consequently, accurately detecting small ship targets in low-resolution images remains a formidable challenge, which actively pursues in the domains of computer vision and remote sensing globally.

Current strategies to address the challenges posed by low resolution and small targets in remote sensing ship images can be broadly classified into two primary approaches. The first approach emphasizes improving the detection accuracy of small targets by enhancing feature extraction capabilities. Chen et al. [2] introduced the super feature pyramid network, which emphasizes small-target features and integrates multiscale information through feature enhancement and aggregation modules. Han et al. [3] integrated a coordinate attention mechanism into the YOLOv5 backbone. They also introduced a small-target detection layer in its neck. This design enhanced positional recognition and reduced interference. Zhang et al. [4] proposed a Taylor finite difference module and a bidirectional attention aggregation module. These modules effectively captured shape features and suppressed noise, improving the accuracy of small-target detection. However, low image resolution still leads to information errors, resulting in false positives (FPs) and missed detections.

Another approach adopts a step-by-step detection method. First, it enhances the clarity of the image before proceeding to detect small ship targets. Xu et al. [5] proposed the YOLO-LMO model tailored for low-resolution marine target detection. The model quantifies image data to 8 bit and incorporates dilated convolutions to distinguish small targets from background features. Musunuri et al. [6] introduced a network based on SRODNet that employs optimized residual blocks to enhance super-resolution techniques. This restores key features in low-resolution images and leverages hierarchical feature extraction to improve detection accuracy. Wang et al. [7] reconstructed high-resolution infrared images from low-resolution inputs, optimizing the training strategy of the YOLOX-S model for improved small-target detection. Jing et al. [8] proposed the YOLOFs algorithm. It combines a feature fusion module, a visual enhancement perception module, and an efficient feature encoding module. These improvements significantly enhance the accuracy and efficiency of detecting

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Fig. 1. Ship detection in low-resolution remote sensing images. First, due to the low image resolution, the feature information of ships tends to become blurred, resulting in missed and false detections. Furthermore, due to the small size of the targets, deeper feature information is prone to loss, especially in capturing key details such as edge textures of small vessels, which further increases the difficulty of detection.

small targets in infrared images. Xu et al. [9] introduced the COCO-Net model, integrating motion information to enhance small-target detection. Using a dual-supervised network, the framework processes data hierarchically by prioritizing target enhancement prior to detection. This approach adaptively optimizes data quality, thereby improving overall performance. Zhang et al. [10] proposed HRFE, which employs a high-resolution feature extractor as a feature-level supervision signal. This guides the model to extract high-resolution features directly from low-resolution images, enhancing the representation of small targets during detection.

The aforementioned methods improve detection accuracy and efficiency for small targets in low-resolution settings. However, they have notable limitation. These methods often lack sufficient diversity when generating high-resolution images, resulting in monotonous and unrealistic details and textures. Additionally, during high-resolution reconstruction, these methods may cause the loss of small-target information, further compromising image quality and subsequent detection accuracy.

Diffusion models (DMs) simulate physical diffusion processes and learn their reverse trajectories to generate realistic, detail-rich images. Such models effectively supplement low-resolution image information, thereby enhancing both image diversity and realism. For instance, Dong et al. [11] introduced a change-aware DM, which separates the denoising process to facilitate efficient super-resolution reconstruction of remote sensing images. This approach improves both reconstruction fidelity and the effectiveness of texture transfer. Similarly, Toker et al. [12] applied a generative image DM to learn the joint data manifold of satellite imagery and their corresponding labels. This approach facilitated the simultaneous generation of high-quality, diverse satellite imagery and corresponding masks, serving as an effective data augmentation strategy. Additionally, Liu et al. [13] enhanced a pretrained YOLO network by extracting feature maps from a DM. This approach achieved remarkable detection performance on low-quality datasets. In small-object detection, Fan et al. [14] combined DMs with implicit neural representations. They improved infrared microtarget detection performance through super-resolution enhancement and multiscale feature alignment. ISTD-diff [15] employs a DM to formulate infrared small-target detection as a denoising-based generative task. A dual-path architecture combined with a channelwise Transformer is utilized to enhance feature representation capabilities, while an attention injection module (AIM) is introduced to optimize the denoising process. Despite the improved detection performance of existing methods utilizing DMs, these approaches often struggle to adapt to small ship targets across different scenarios. Moreover, due to the minimal size of small targets in images, noise in low-quality remote sensing images further hampers the detection of these small targets. In existing detection frameworks, spatial information is often diluted during global modeling. This leads to blurred edges and loss of details for small ship targets, reducing detection performance. The advantages and weaknesses of the current method in addressing this type of problem are summarized in Table I.

As summarized in Table I, existing approaches exhibit distinct strengths in enhancing specific capabilities. Some methods demonstrate strong performance in texture reconstruction and spatial-edge representation, making them well-suited for scenarios with rich texture variations such as remote sensing or natural scenes. Others focus more on feature fusion and multiscale adaptation, showing advantages in detecting complex targets across varying scales. However, certain limitations remain, including incomplete detailed restoration and limited cross-scenario generalization. These drawbacks may lead to performance instability under challenging conditions such as varying illumination or complex backgrounds. This suggests that while notable progress has been made for specific tasks, further efforts are needed to improve adaptability and robustness across diverse application scenarios.

In conclusion, small ship target detection in low-resolution remote sensing images remains a challenging task, primarily due to two factors: 1) limited image clarity, which hampers the visibility of ship features and leads to false detections or missed targets and 2) the extremely small size of ships, which results in the degradation of deep semantic features and weakens the expression of fine structures such as edges and textures. Therefore, this article proposes a small-target ship detection model for low-resolution remote sensing images based on a DM (LRS^2-DM). The core contributions of this article are as follows.

1) To address the blurred ship features caused by low-resolution remote sensing images, we propose cognitive conditions as inputs and integrate a low-level super-resolution (L^{2}SR) module to enhance image

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TABLE I

COMPARATIVE ANALYSIS OF REPRESENTATIVE METHODS FOR SMALL SHIP DETECTION IN LOW-RESOLUTION REMOTE SENSING IMAGES. THIS TABLE SUMMARIZES THE MAIN ADVANTAGES AND LIMITATIONS OF SEVERAL STATE-OF-THE-ART APPROACHES, INCLUDING THEIR PERFORMANCE IN ENHANCING SMALL-OBJECT DETECTION, UTILIZING MULTISCALE FEATURE FUSION, IMPROVING SPATIAL-EDGE REPRESENTATION, ENRICHING TEXTURE AND DIVERSITY, AND ADDRESSING ISSUES LIKE INCOMPLETE DETAILED RESTORATION AND LIMITED CROSS-SCENARIO GENERALIZATION

Advantages & Weaknesses	Chen et al. [2]	Han et al. [3]	Musunuri et al. [6]	Xu et al. [9]	Dong et al. [10]	ISTD-diff. [15]
Small object detection enhancement	✓	✓	✓	✓	✓	✓
Multi-scale feature fusion	✓	✓				✓
Spatial–edge representation enhancement					✓	✓
Texture and diversity improvement	✓		✓		✓	
Incomplete detail restoration	✓	✓				✓
Poor cross-scenario generalization		✓	✓	✓	✓	✓

clarity. Essential visual features for the detection task are fed into the network. The DM integrates textual information to generate task-specific prior knowledge, effectively enhancing image resolution.

2) To tackle the issue of missing texture and detailed information in small targets, we introduce a spatial refinement module (SRM). This module aims to compensate for the loss of ship information in remote sensing images through spatial structure transfer, refining minute features and improving the detection accuracy of small targets.
3) Finally, we advance an optimized loss module to enhance network integration and mitigate the negative impact of noise introduced by the DM in remote sensing images. Moreover, the loss function for object detection is redefined to adapt to tasks at different levels, ensuring enhanced model stability.

The remainder of this article is structured as follows. Section II reviews related works on low-resolution and small-object detection in remote sensing images. Section III describes the proposed LRS^2-DM method in detail. It covers cognitive conditional input (CCI), the L^{2}SR module, the SRM, and the loss function. Section IV presents the experimental study. It introduces the experimental setup, datasets, and evaluation metrics, followed by an analysis of the results. Finally, Section V concludes this article and discusses future research directions.

II. RELATED WORK

With the continuous advancement of deep learning technology, image enhancement and small-object detection algorithms have shown significant potential in the field of computer vision. Image enhancement techniques focus on improving image quality, which directly improves the performance of various visual tasks, particularly small-scale object detection. At the same time, small-object detection algorithms tackle the complex challenges of detecting small-scale targets, driving rapid advancements in remote sensing image detection technology.

A. Image Enhancement Algorithm

Traditional image enhancement methods, such as histogram equalization [16], edge detection [17] (e.g., the Canny and Sobel operators), and linear transformations [18], are relatively simple and intuitive. These methods rely on straightforward principles and do not require complex model training, making them easier to understand and implement. However, compared to deep learning-based algorithms, traditional methods exhibit limited generalization capabilities and suboptimal performance in enhancing low-resolution targets in remote sensing images.

Deep learning-based image enhancement algorithms primarily include convolutional neural networks (CNNs), generative adversarial networks (GANs), and DMs. Pang et al. [19] developed a dual-layer deep CNN that extracts and enhances thermal radiation features while preserving structural integrity, achieving high-quality infrared image enhancement. Jeon et al. [20] leverages high-quality information from adjacent multiview images as references, integrating optical flow estimation, image warping, and synthesis techniques to enhance target image quality. Yin et al. [21] employs a generator with dual discriminators to generate fused images using pyramid decomposition, residual attention weighting, and cross-scale pyramid attention mechanisms, effectively enhancing the overall semantic information within the scene. Sun et al. [22] integrate image appearance and linguistic understanding to facilitate high-quality reference image generation, restoring semantic and realistic detailed features through a dedicated integration module. An et al. [23] utilize a cross-attention encoder–decoder architecture to effectively incorporate conditional information, which further enhances image quality by leveraging accelerated sampling techniques from DDIMs.

To address noise, texture loss, and checkerboard artifacts caused by low contrast and complex textures in remote sensing images, Han et al. [24] proposed an inverse feature fusion super-resolution network. This approach integrates high- and low-level semantics to improve global performance and incorporates a feature expansion module to enhance local details. Liu et al. [25] introduced an edge enhancement module, optimized the mask branch, and applied subpixel convolution. These improvements enhanced edge information quality and strengthened image consistency. Han et al. [26] developed an efficient hybrid conditional DM and incorporated Fourier high-frequency spatial constraints. This method optimized high-frequency information recovery in the reverse diffusion process, leading to improved detailed reconstruction.

However, CNNs and GANs often yield unrealistic results in remote sensing target enhancement tasks, including blurred details, unnatural textures, or significant discrepancies with the actual scene. In recent years, super-resolution and object

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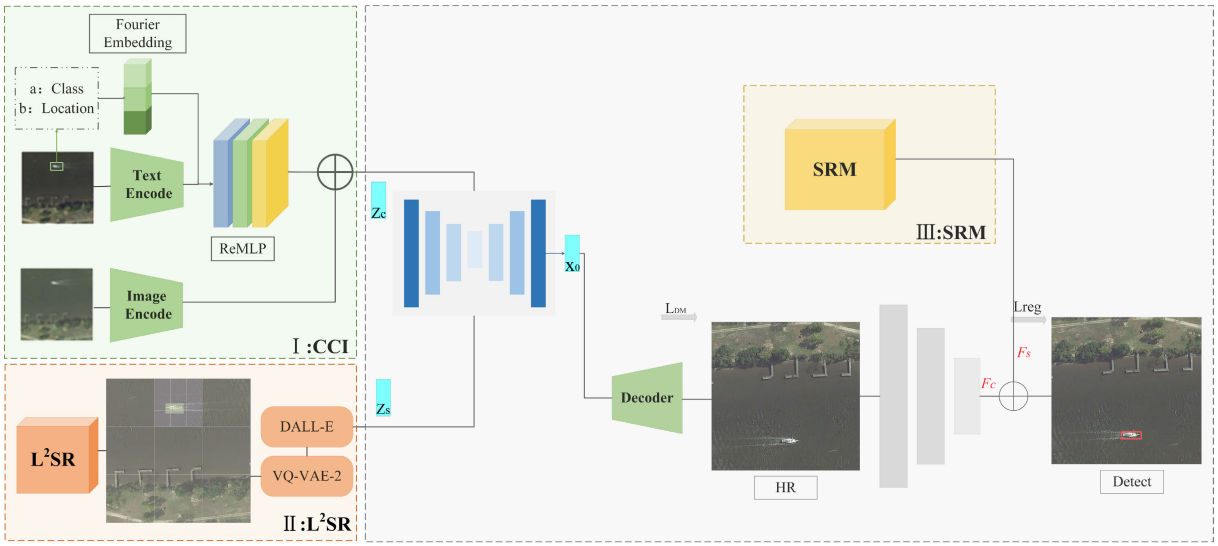


Fig. 2. Overall framework of the LRS^2-DM. First, high-quality remote sensing images are generated by integrating the CCI with the L^{2}SR module. In the subsequent detection stage, the SRM is further introduced to enhance feature representation, thereby improving object detection performance.

detection algorithms based on DMs have shown remarkable advantages in the field of image processing [27], [28], [29]. The unique reverse diffusion process allows for the stepwise generation of high-quality images from noise, transforming originally blurred or unrecognizable targets into clearer and more distinguishable forms.

B. Small-Object Detection Algorithm

In the field of object detection, ship targets in remote sensing images typically occupy fewer than 32 × 32 pixels [30], categorizing them as small targets. The shape, texture, and position of small targets are frequently blurred, posing significant challenges for feature extraction. Consequently, effective feature extraction for small targets in remote sensing images has become a pivotal challenge in object detection.

Common small-target detection algorithms often rely on feature pyramids [31], which integrate internal and cross-scale features within the latent space. DSFPAP-Net [32] incorporates 3-D information, allowing the model to autonomously learn spatial features. It accelerates convergence via an improved loss function, thereby enhancing detection accuracy for small targets. Zhang et al. [33] integrated a segmentation model and introduced a novel query design to encode multiscale features of small targets, enhancing detection performance. Jing et al. [8] improved detection precision by integrating feature fusion, visual perception, and feature encoding. To address localization challenges. Tian et al. [34] introduced a target-feedback small-target linkage loss function, dynamically adjusting focus based on object size to enhance small-target detection. To ensure detector accuracy and simplicity, Zhang et al. [35] proposed IRprune, which optimizes the model through channel pruning and wavelet structure regularization. CourtNet [36] enhances recall and accuracy through three sequential networks, which weight the results to balance precision and recall. For data augmentation, IRSAM [37] enhances edge features and suppresses noise via the Perona–Malik diffusion module, while the granularity-aware decoder fuses multiscale information to boost accuracy and reduce background interference. DetDiffusion [38] introduces visual attributes via segmentation and employs a DM, which generates high-quality images to effectively enrich the training dataset.

Small-target detection algorithms represent a critical domain in computer vision, focusing on the precise recognition and localization of small targets in images or videos. These methods employ various technical approaches to improve detection performance. Nevertheless, small-object detection algorithms still struggle with challenges such as limited localization accuracy, low image resolution, and weak generalization capabilities.

III. Methods

This study focuses on addressing the challenge of small-object detection in low-resolution remote sensing imagery. To mitigate resolution limitations, the method incorporates cognitive-conditioned input and an L^{2}SR module to generate reference images as auxiliary guidance. This enhances the quality of the super-resolved outputs. Additionally, to compensate for the loss of fine details in small targets, an SRM is employed. This module sharpens object-level features and improves the accuracy of ship detection in remote sensing scenes.

This article begins by introducing the dual-branch image generation framework illustrated in Fig. 2, focusing on the cognitive-conditioned input described in Section III-A1. It, then, elaborates on the L^{2}SR module (Section III-A2), which leverages the DM's capability to synthesize image content from textual priors. Section III-B presents the SRM. Finally, Section III-C describes the enhanced loss module designed to improve training stability and detection robustness.

A. Dual-Branch Image Generation Module

DMs [39] are a class of generative models grounded in probability theory, as illustrated in Fig. 3. In the forward diffusion process, starting with an initial image x_0, the model incrementally applies Gaussian noise at each time step $t \in$

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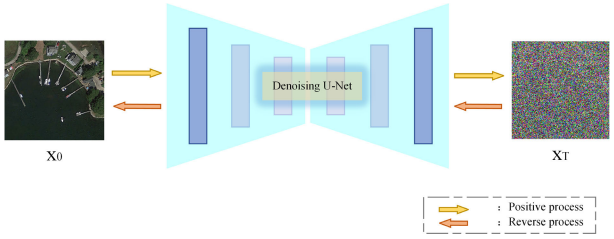


Fig. 3. Workflow of the DM. DMs progressively corrupt images into noise through a forward process and gradually reconstruct them from noise via a reverse process, learning to denoise effectively during training.

$\{0, 1, \ldots, T\}$, resulting in a fully noisy image by the end of the process. During the reverse diffusion process, the DM iteratively removes noise from standard Gaussian noise x_T to estimate a clear x_0 and reconstruct the original image, as shown in the following equations:

$$x_t = \sqrt{\bar{a}_t} x_0 + \sqrt{1 - \bar{a}_t} \epsilon_t \tag{1}$$

$$L = \mathbb{E}_{x_0, \epsilon \sim \mathcal{N}(0,1)} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]. \tag{2}$$

Here, $\bar{a}_t$ represents a predefined scheduling variable, $\epsilon_t \sim \mathcal{N}(0, 1)$ denotes the added Gaussian noise, ϵ_θ corresponds to the output of the pretrained noise network, and ϵ_t serves as the supervised noise.

To enhance the reconstruction and localization of small objects in low-resolution remote sensing imagery, this study introduces a semantic-guided module called CCI and a structure-guided module termed L^{2}SR. These two modules jointly serve as conditional inputs to the DM. The CCI module integrates class labels and positional information. Through positional encoding and multimodal fusion, it produces a semantic guidance vector, Z_c, which refines the spatial distribution of small targets during the diffusion-based reconstruction. This enhances the model's focus on small-object regions. Meanwhile, the L^{2}SR module performs an initial structural restoration of the input image, extracting structural features, Z_s, that preserve edge and contour details. During the reverse denoising process of the DM, Z_c and Z_s are embedded as conditional inputs at each time step. They guide the progressive reconstruction of the target image from the noisy observation x_t, as formulated in the following:

$$\hat{x}_0 = \varepsilon_\theta(x_t | Z_s, Z_c). \tag{3}$$

Guided by both structural and semantic constraints, the diffusion process not only improves overall image clarity but also substantially enhances the restoration and detection of small-object regions.

1) Cognitive Condition Input: To tackle the challenge of capturing small-object features in low-resolution ship images obtained via remote sensing technology, this study introduces CCI into the generative framework. This methodology enhances the generation of accurate small-target ship images, which provides reliable data support for downstream ship detection tasks.

The cognitive conditions include key visual features critical for accurate target detection, such as bounding boxes and categories, which are seamlessly integrated into the training process of the generative model. In this process, the model leverages the pretrained contrastive language-image pretraining (CLIP) architecture and employs a text encoder to integrate prior image categories alongside supplementary contextual conditions. This approach facilitates precise extraction and reproduction of details in remote sensing images, which results in highly accurate imagery aligned with the expected objectives.

The proposed framework exploits the pretrained knowledge within DMs, generating prior information regarding categories and bounding boxes through the CLIP encoder. The fundamental input is formulated as the classes and bounding boxes of M objects in the image, as expressed in the following equations:

$$C_{\text{lab}} = \left[c_1, \ldots, c_M\right] \tag{4}$$

$$L_{\text{loc}} = \left[l_1, \ldots, l_M\right]. \tag{5}$$

Here, c_i denotes a specific class from the set of training classes C, while l_i represents the bounding box associated with the ith instance, defined by normalized coordinate values $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$.

Specifically, for each image x, a detector such as Faster R-CNN [40] or DETR [41] is employed. The detector generates n predicted bounding boxes, denoted as $\mathbf{d} = [d_1, d_2, \ldots, d_n] = D(x)$. All bounding boxes in $D(x)$ are normalized with respect to their coordinates and considered as cognitive attributes. In this framework, each predicted bounding box for a target is characterized by two attributes: category (c_i) and location (l_i). The classification labels $\mathbf{C}_{\text{lab}}$ and their corresponding target locations $\mathbf{L}_{\text{loc}}$ are selected for all detected instances. Its cognitive condition input is defined as a set of label-location pairs, as shown in the following equation:

$$\{\mathbf{C}_{\text{lab}}, \mathbf{L}_{\text{loc}}\} = \{(c_1, l_1), (c_2, l_2), \ldots, (c_n, l_n)\}. \tag{6}$$

Next, the specific label "$\mathbf{C}_{\text{lab}}$" is processed through the text encoder of CLIP, enabling it to accomplish a text-to-image conversion. Simultaneously, the reparameterized multilayer perceptron (ReMLP) proposed by Ding et al. [42] is utilized, to enhance the fusion of image feature vectors and improve the accuracy of image recognition.

This module leverages cognitive attributes to design effective textual prompts by incorporating category labels and location information as their foundation. Specifically, the cognitive prompt is formulated as "an image with target category." Here, the target object is denoted as $[(c_1, l_1), (c_2, l_2), \ldots, (c_n, l_n)]$, and n represents the total number of predicted bounding boxes. The comprehensive attribute set and effective prompt text provide a more thorough understanding of the characteristics of each target object. These detailed descriptions effectively capture multiple aspects and features of the targets, furnishing a richer and more comprehensive informational foundation for cognitive conditions.

This method ensures that the obtained remote sensing images possess both a high degree of realism and compliance with the basic principles of visual cognition. Meanwhile, preprocessing based on cognitive conditions can effectively reduce the computational load of the model. This significantly improves the accuracy and reliability of detection and enhances the efficiency of the model.

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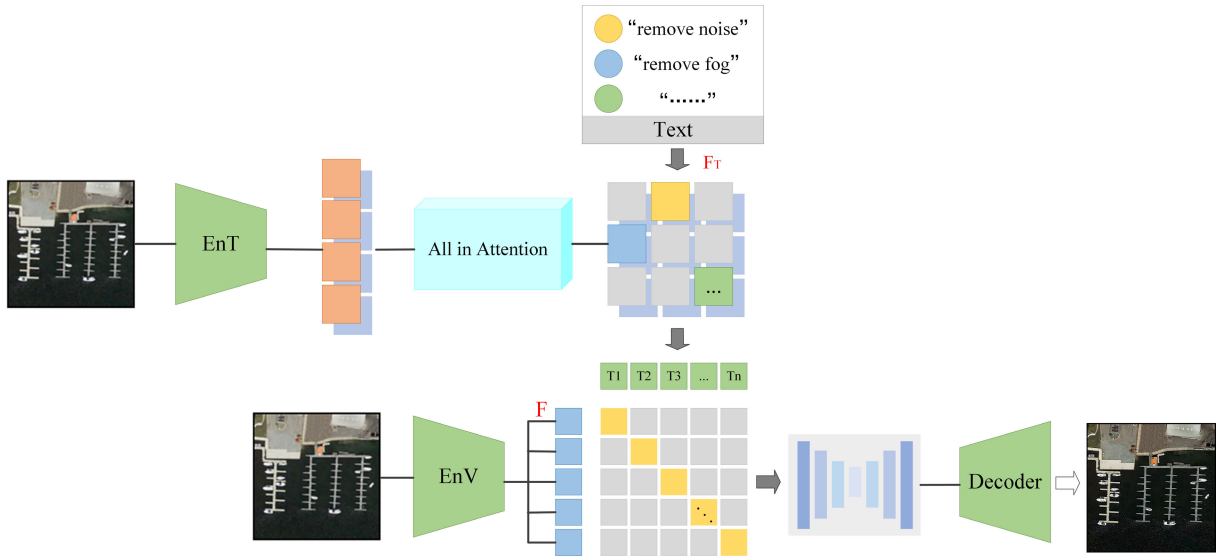


Fig. 4. Schematic of the L^{2}SR module. Flowchart of the proposed module: the input image is processed through the CLIP encoder to extract features, followed by text prompt generation using the all-in-attention mechanism. The SMFA module enhances feature modeling, and the target image is generated by combining the encoded image with the text prompt. The image is then refined using DALL-E, VQ-VAE-2, and U-Net for high-quality output.

2) L^{2}SR Module: Ship targets in low-resolution remote sensing images appear blurry and lack discernible details, limiting the effective identification of shape and texture features. Therefore, enhancing image resolution significantly improves image details and visual features, making the distinction between ships and the background more pronounced. This enables more accurate ship detection and captures additional remote sensing information.

Models such as GPT [43] and BERT [44] demonstrate zero-shot capabilities, allowing them to execute tasks without explicit training for specific objectives. These models possess strong semantic learning abilities, allowing them to handle low-level visual processing without compromising their core functionalities. Similarly, DMs demonstrate exceptional zero-shot learning capabilities. They can capture complex visual patterns and structures without relying on tedious task-specific training for particular image styles, scenes, or objects. This capability allows them to generate diverse visual outputs. However, the stochastic nature of DMs may result in unintended alterations to remote sensing images. To address this issue, we propose a model named L^{2}SR, as shown in Fig. 4.

This module serves a crucial role by delivering prior knowledge. It is tailored specifically for image processing tasks. This ensures that the generated reference images are precisely aligned with the textual and visual content requirements of the task. This ensures that the generated reference images are precisely aligned with the textual and visual content requirements of the task. Specifically, for a given input remote sensing image I, basic features are initially extracted through a CLIP-based visual encoder EnT(I). Subsequently, the all-in-attention [22] mechanism is applied to generate the textual prompt F_T in the following equation:

$$F_T = \text{AiA}(\text{EnT}(I)). \tag{7}$$

The all-in-attention module effectively integrates low-resolution attention, reference attention, and cross-attention, enabling dynamic switching of task conditions and multimodal information fusion to support multitask visual understanding in complex scenarios. However, small-object ship detection in low-resolution remote sensing images still faces challenges, such as missing local textures and feature blurring. To address this, we introduce a self-modulated feature aggregation (SMFA) [45] module to enhance the model's ability to jointly model both local and nonlocal information. MFA consists of a lightweight approximate self-attention (EASA) branch and a local detail enhancement (LDE) branch. The former captures long-range dependencies, while the latter compensates for missing edge information in small targets. Additionally, a partially convolutional feedforward network (PCFN) is introduced to further refine the fused features, improving their discriminative power and robustness. Fig. 5 illustrates the architecture after fusion. This design not only preserves the original multimodal task control capabilities, but also significantly enhances small-object detection performance in low-resolution images, while effectively mitigating the issue of information drift during the DM guidance.

This module utilizes the image encoder EnI(I) to deeply encode the input image I, thereby capturing its intrinsic informational representation F. Subsequently, based on the provided textual prompt F_T, which is enriched with descriptions and guidance for the desired image content. This textual condition is then integrated with the encoded image features to construct the target image I_F, enriched with additional informational characteristics, as represented in the following equation:

$$F_{(I_F)} = \text{fusion}\left(F, F_T, F^t\right). \tag{8}$$

Here, F^t denotes the embedded time, with t representing the time step in the diffusion process.

The generated reference images act as effective visual guidance for subsequent tasks. In this article, DALL-E [46] is used as a tool capable of not only understanding text descriptions but also converting them into corresponding image

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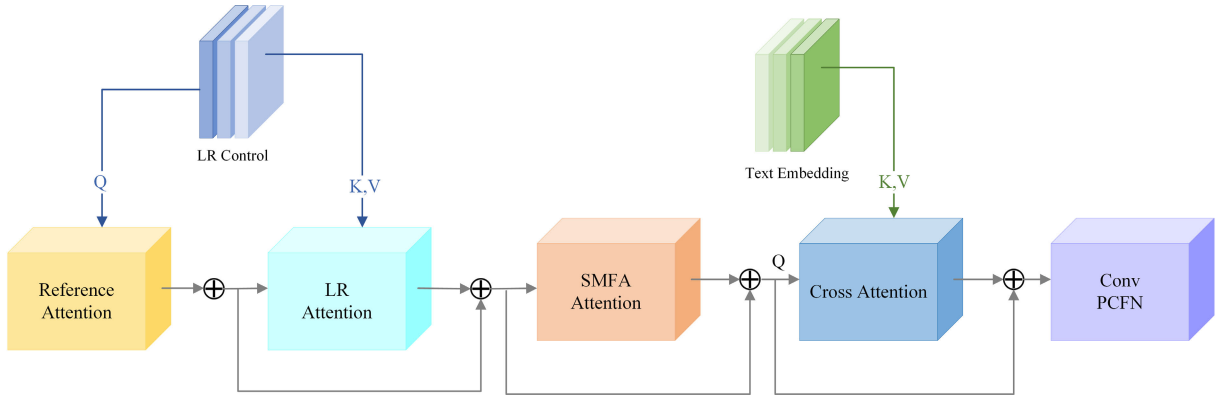


Fig. 5. Schematic of the SRM. Diagram of the SMFA module: it combines EASA for long-range dependencies, LDE for small-target edge enhancement, and PCFN for refined feature fusion, improving detection accuracy and robustness in low-resolution images.

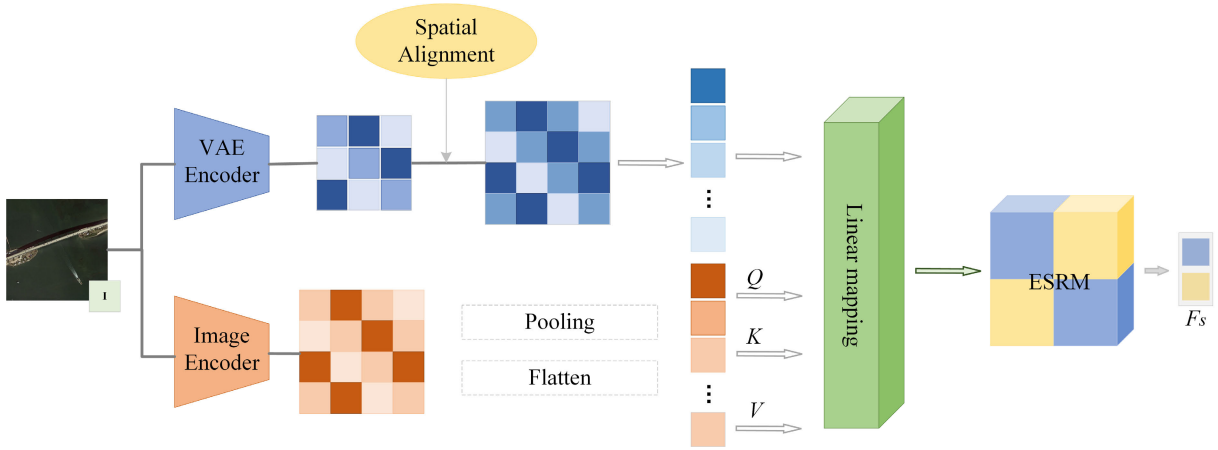


Fig. 6. Schematic of the SRM. Diagram of the proposed SRM. The VAE encoder captures key features, followed by adaptive pooling for spatial alignment. SRM enhances small-target features from multiple spatial layers, improving detection accuracy. The ESRM uses multihead attention to further strengthen feature representation and robustness in complex backgrounds.

representations, enabling sophisticated text-to-image feature representation. Subsequently, these representations undergo further processing through VQ-VAE-2 [47], which learns high-level semantic information from the images, facilitating the generation of fine-grained control features across multiple scales. Finally, these control features are incorporated into a U-Net architecture, which leverages its capabilities for contextual information fusion and detailed restoration to produce high-quality remote sensing images.

B. Spatial Refinement Module

Due to the inherently small size of ship targets in remote sensing images, it remains challenging to fully address the loss of edge and texture information caused by their limited scale. This limitation further degrades fine-grained image details, which negatively impacts the precise extraction and recognition of target features. To better support detection tasks during image generation, we propose an SRM, as depicted in Fig. 6. This module is specifically designed to tackle the challenges posed by information loss due to small-target sizes, thereby enhancing the accuracy and completeness of image detection.

Specifically, the VAE encoder is initially utilized to extract key information from the input image I. Subsequently, an adaptive average pooling layer is applied to spatially align the feature maps, ensuring uniform spatial dimensions. This process results in spatially aligned input feature maps, denoted as x. However, due to their limited size, small targets are frequently overshadowed by extensive background information in complex scenes, which hinders the direct extraction of their unique and distinguishable features. To address this issue, SRM is utilized, and the refined features produced by SRM are expressed as shown in the following equation:

$$\mathrm{SRM}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V. \tag{9}$$

Here, Q is the query vector, K is the key vector, V represents the value vector, d_k is the dimension of the key vector, and T denotes the transpose operation.

The SRM extracts and amplifies feature representations of small targets from diverse spatial levels and perspectives, thereby improving their recognizability and detection accuracy within complex backgrounds. To further improve the feature representation of small targets, a multihead attention mechanism inspired by [48] is incorporated. Accordingly, we redefine the enhanced SRM (ESRM) as follows:

$$\mathrm{ESRM}(Q, K, V) = \mathrm{concat}\left(h^1, h^2, \ldots, h^j\right) \tag{10}$$

$$h^j = \mathrm{Head}^j = \mathrm{SRM}\left(Q_j, K_j, V_j\right). \tag{11}$$

Here, j denotes the number of attention heads. The formula for calculating spatial refinement features F_s is expressed as

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follows:

$$F_s = \mathrm{ESRM}(x, x'). \tag{12}$$

Here, x' acts as a unique query, designed to extract or focus on specific information from the feature map, also denoted as x. This approach captures the features of small targets across multiple spatial levels or image scales. It enhances the robustness and generalization capacity of small-target features. This allows the model to adapt more effectively to small-target detection and recognition tasks under diverse scene and lighting conditions.

C. Loss Function

The inherent uncertainty in remote sensing images makes them vulnerable to noise during random diffusion, which degrades image quality and analytical accuracy. To address this, we adopt the stable DM with a distribution stability index as the framework for image generation. The loss function L_{DM} is defined as follows:

$$L_{\mathrm{DM}} = \mathbb{E}_{x_0, f, \varepsilon \sim \mathcal{N}(0,1)} \left\| \varepsilon - \varepsilon_\theta \left(x_t, t, \tau_{\theta_1}(y_1) + \tau_{\theta_2}(y_2) \right) \right\|^2. \tag{13}$$

Here, $x_t = (\bar{a}_t)^{1/2} x_0 + (1 - \bar{a}_t)^{1/2} \varepsilon_t$ represents the denoising process at time step t, while x_0 denotes the representation of the original ground-truth image. ε_θ denotes the predicted output of the neural network. $\tau_{\theta_1}(y_1)$ and $\tau_{\theta_2}(y_2)$ represent the input regulation of cognitive conditions and L^2SR modules, respectively. This function considers the reconstruction accuracy of the image content. Additionally, it incorporates the distribution stability index as a regularization term. This design enhances the stability and consistency of image features throughout the diffusion process. The L_{DM} framework precisely regulates the mapping relationship between the data space and the latent space, which effectively reduces the adverse effects of noise on image generation. This ensures that the generated images exhibit high clarity and rich detail. In remote sensing data, outliers frequently occur due to environmental interference or data transmission errors, which creates a significant challenge. Moreover, gradient vanishing during model training often impedes effective learning, which further complicates the optimization process. To address these challenges, the mean absolute error (MAE) is adopted as the regression loss function for its robustness to outliers, as shown in the following equation:

$$L_{\mathrm{reg}} = \frac{1}{m} \sum_{i=1}^{m} \left| y_i - \mathrm{SRM}(y_{\mathrm{pre}}) \right|. \tag{14}$$

Here, y_i represents the ground-truth value of the ith sample, y_{pre} denotes the predicted value of the ith sample, and m is the number of samples.

Therefore, the overall objective function is formulated as shown in the following equation:

$$L = \begin{cases} \alpha L_{\mathrm{DM}} + \beta L_{\mathrm{reg}}, & \text{for detection} \\ L_{\mathrm{DM}}, & \text{for diffusion.} \end{cases} \tag{15}$$

Here, α and β are the loss weights for the diffusion loss L_{DM} and the regression loss L_{reg}, respectively.

IV. Experiments

A. Experimental Setup

The LRS^2-DM method in this study is implemented on the PyTorch deep learning framework running on Ubuntu v20.04. The hardware setup included an Intel Core i7-10700F CPU, an NVIDIA GeForce RTX 3070 GPU (8-GB VRAM), and 32 GB of RAM. The method utilized ResNet-50 as the pretrained backbone network. The network pretrained on ImageNet was employed to analyze the feature distribution of the generated images. The input image size was resized to 512×512, and the batch size was set to 8. The network was trained for a total of 300 epochs. Specifically, to effectively mitigate false convergence in small-target detection and reduce sample misclassification, the intersection over union (IoU) threshold was set to 0.3 during training and 0.5 during detection. During the image generation process, the stable DM [49] was employed. The AdamW [50] optimizer was utilized for optimization.

B. Dataset

To evaluate the performance of this method, we used the publicly available optical remote sensing datasets DOTA-v2.0-Ship and S-Ship. Robustness tests were also conducted on the self-constructed dataset Ship-Own. As shown in Fig. 7, representative sample images from the three datasets are presented. To further verify the effectiveness of the proposed method under low-resolution conditions, all images were downsampled to resolutions above 1 m to simulate low-resolution scenarios. The source code and relevant resources will be progressively organized and made publicly available at: https://github.com/shangxuepang/LRS2-DM.git. Further updates and improvements will be released through this repository in due course.

1) DOTA-v2.0-Ship: DOTA [51] is a large-scale remote sensing image dataset derived from multiple sensors, including Google Earth, GF-2 satellites, and aerial imagery. Compared to v1.0, v2.0 enhances annotations for extremely small targets (fewer than ten pixels) while maintaining the original imagery. This study focuses on detecting small ship targets. Therefore, 420 ship-containing images were selected from the DOTA dataset to create the DOTA-v2.0-Ship dataset, comprising a total of 36 258 ship targets. To further validate the algorithm's accuracy, multiscale downsampling was applied to the original images. These were, then, segmented into 1024×1024 patches with a 20% overlap, increasing the training samples to 1978 and the testing samples to 548. The total number of ship targets increased to 54 796.

2) S-Ship: This dataset is a combination of the DIOR [52] and HRRSD [53] datasets, with imagery primarily sourced from the Google Earth sensor. We extracted ship images from both datasets and divided them into three subsets: training, testing, and validation sets. Specifically, the training set contains 1600 ship images, the testing set includes 3388 images, and the validation set comprises 1600 images. However, these ship images do not exclusively focus on small ship targets but include vessels of various sizes. Subsequently, we filtered suitable small ship target images from the three subsets. Due to

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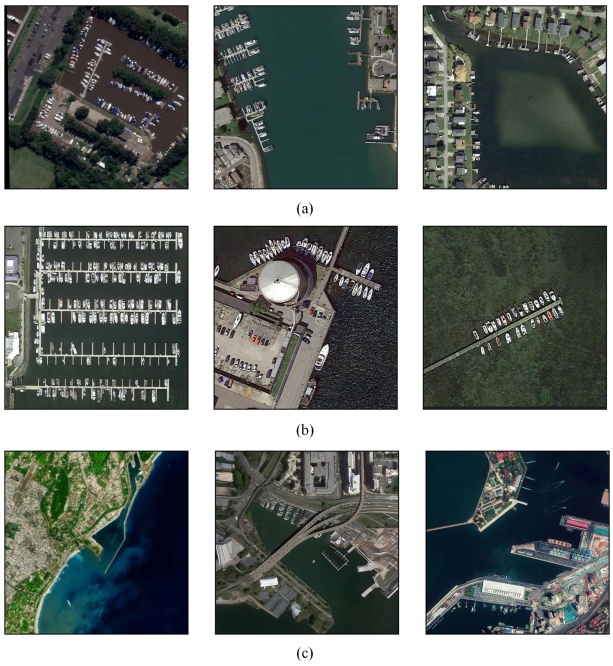


Fig. 7. Sample images from the Ship-Own dataset. Specifically, (a) corresponds to the DOTA-v2.0-Ship dataset, (b) corresponds to the S-Ship dataset, and (c) corresponds to the Ship-Own dataset.

the large size of the original images, they were segmented into multiple 1024 × 1024 patches containing ships, with a 20% overlap, forming the S-Ship dataset. In this refined dataset, the training set consists of 1636 images, while the testing set includes 641 images. This ensures adequate and effective support for small ship target detection tasks.

3) Ship-Own: This study focuses on the detection of small ship targets in low-resolution remote sensing imagery and introduces a custom dataset named *Ship-Own*. The dataset features diverse sources, including Google Earth,[1] Jilin-1 satellite imagery,[2], and other open remote sensing platforms. Specifically, the dataset comprises 1254 images from Google Earth, 816 images from the Jilin-1 satellite, and 438 images from additional sources. All raw images were first preprocessed and filtered to remove low-quality or irrelevant content. Subsequently, ship targets were manually annotated using the open-source tool LabelImg, ensuring high annotation accuracy and consistency. To facilitate model training, the annotated images were uniformly cropped into patches of 1024 × 1024 pixels. The dataset was then split into training, validation, and test subsets following a 7:2:1 ratio. The distribution of ship sizes within the Ship-Own dataset is illustrated in Fig. 8.

C. Evaluation Metrics

In the task of ship target detection, we adopted the widely recognized average precision (AP) as the core metric for performance evaluation, which measures the effectiveness of different methods. A predicted bounding box is classified as a true positive (TP) if its IoU exceeds a predefined threshold. Otherwise, it is considered an FP. If no predicted bounding box covers the target area, it is labeled as an FP. Otherwise,

[1]https://earth.google.com/
[2]https://www.jl1mall.com/

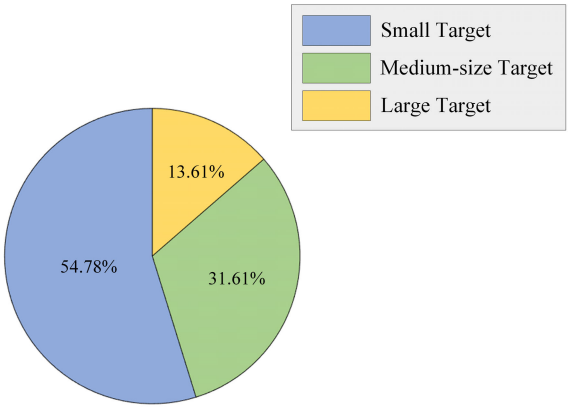


Fig. 8. Sample images from the Ship-Own dataset. Small-sized ship targets account for 54.78%, medium-sized ships make up 31.61%, and large-sized targets constitute 13.61%.

the area is considered a true negative (TN). Based on these classifications, precision and recall are further calculated to comprehensively evaluate the performance of the detection methods, as shown in the following equations:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{17}$$

To provide a more intuitive reflection of performance, AP is defined as follows:

$$\text{AP} = \int_0^1 P(R)\, dR. \tag{18}$$

Here, $P(R)$ represents the precision–recall (P–R) curve.

As our research focuses on ship detection technology for small targets, we conducted a series of robustness experiments using the Ship-Own dataset and performed a comprehensive analysis using multiple evaluation metrics. Specifically, our analysis focuses on AP at three different scales: small-scale AP (APS), medium-scale AP (APM), and large-scale AP (APL).

The formulas for AP(s), AP(m), and AP(l) are

$$\text{AP}(s) = \text{AP}, \quad \text{area} \le 32 \times 32 \tag{19}$$

$$\text{AP}(m) = \text{AP}, \quad 32 \times 32 < \text{area} \le 96 \times 96 \tag{20}$$

$$\text{AP}(l) = \text{AP}, \quad 96 \times 96 < \text{area} \tag{21}$$

where area denotes the pixel count covered by the ship.

To validate the effectiveness of the L^{2}SR module, two key image quality evaluation metrics were adopted: peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM). These metrics comprehensively and accurately assess the differences and structural similarity between the reconstructed and original images. They ensure that the module improves resolution while preserving image details and structural integrity. The evaluation metrics for SSIM and PSNR are defined as shown in the following equations:

$$\text{PSNR} = 10 \log_{10}\left(\frac{\text{MAX}_I}{\text{mse}}\right) \tag{22}$$

$$\text{SSIM}(a, b) = \frac{(2\mu_a\mu_b + c_1)(2\sigma_{ab} + c_2)}{(\mu_a^2 + \mu_b^2 + c_1)(\sigma_a^2 + \sigma_b^2 + c_2)}. \tag{23}$$

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TABLE II

COMPARISON OF DIFFERENT ALGORITHMS ON THE DOTA-V2.0-SHIP DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	AP (%)	Recall	FLOPs(G)	Paras(M)	FPS
Faster R-CNN	78.14	76.85	113.5	38.74	7.5
COCO-Net	89.56	89.95	20.7	**11.26**	31.4
LMO-YOLO	90.36	90.46	25.4	12.24	34.5
DiffusionDet	88.21	91.32	45.3	43.62	32.3
HFPNet	93.15	93.66	25.6	16.75	**50.8**
DCDet	87.34	88.31	22.7	45.25	39.6
YOLO-SM	94.42	93.24	**13.3**	12.42	35.3
Ours	**95.34**	**95.21**	24.3	35.08	49.2

Here, MAX denotes the maximum pixel value of image I, and mse represents the mean squared error between the reconstructed image and the reference image. μ_a and μ_b represent the mean luminance values of the two images, while σ_a^2 and σ_b^2 denote their contrasts. σ_{ab} is the covariance between the two image patches, used to evaluate their structural similarity. The constant c_i ensures numerical stability by preventing the denominator from becoming zero.

D. Comparative Experiments

To demonstrate the effectiveness of the proposed LRS^2-DM method, we plan to compare it against six other target detection algorithms: Faster R-CNN [54], COCO-Net [8], LMO-YOLO [4], DiffusionDet [55], HFPNet [2], DCDet [56], and YOLO-SM [57]. The comparative detection performance of these methods on the DOTA-v2.0-Ship dataset is summarized in Table II.

As shown in Table II, our proposed algorithm demonstrates notable advantages across all performance metrics. Specifically, after processing through the L^{2}SR and SRMs, the algorithm achieves the highest AP and recall. Compared to the top-performing YOLO-SM algorithm, our method achieves a 0.92% improvement in AP. Although the algorithm has relatively higher FLOPs and Paras, it maintains a high frame-per-second (FPS) rate through preprocessed cognitive inputs. The FPS difference with the fastest HFPNet algorithm is only 1.6. Notably, while the YOLO-SM algorithm achieves minimal FLOPs, its FPS performance lags behind our method by a significant margin of 13.9. Additionally, compared to the diffusion algorithm of DiffusionDet, our method achieves a higher AP value.

To visually demonstrate and compare the detection performance of various algorithms, we performed a visualization analysis, with the results shown in Fig. 9. Arranged in the ascending order of AP values, Fig. 9(a)–(g) corresponds to the visualization of detection results for DCNet, DiffusionDet, COCO-Net, LMO-YOLO, HFPNet, YOLO-SM, and ours, respectively. Among these, the other algorithms exhibit varying degrees of missed detections and FPs.

In summary, the LRS^2-DM algorithm demonstrates strong performance on the DOTA-v2.0-Ship dataset. It achieves high accuracy for small ship targets in low-resolution images while maintaining a good balance among computational complexity, model size, and processing speed.

TABLE III

COMPARISON OF DIFFERENT ALGORITHMS ON THE S-SHIP DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	AP (%)	Recall	FLOPs(G) ±0.1%	Paras(M) ±0.1%	FPS ±3.0%∼10.0%
Faster R-CNN	78.25	77.95	113.5	38.74	7.5
COCO-Net	89.72	89.74	20.7	**11.26**	31.4
LMO-YOLO	90.26	91.46	25.4	12.24	34.5
DiffusionDet	89.19	91.34	45.3	43.62	32.3
HFPNet	93.66	94.44	25.6	16.75	**50.8**
DCDet	88.64	89.26	22.7	45.25	39.6
YOLO-SM	94.41	94.29	**13.3**	12.42	35.3
Ours	**96.12**	**96.87**	24.3	35.08	49.2

TABLE IV

COMPARISON OF ALGORITHMS ON THE SHIP-OWN DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Method	mAP	AP(s)	AP(m)	AP(l)
COCO-Net	87.24	86.72	88.31	90.12
LMO-YOLO	88.69	88.15	89.91	91.25
DiffusionDet	85.32	84.22	86.46	87.53
HFPNet	90.64	89.79	93.74	94.24
YOLO-SM	91.68	89.30	**94.59**	**96.35**
Ours	**92.14**	**91.95**	93.62	95.11

To further validate the feasibility of the algorithm, we conducted comparative experiments on the S-Ship dataset. The detection results are presented in Table III. Additionally, the detection results were visualized, as shown in Fig. 10.

Based on the analysis of Table III, the proposed algorithm achieves the highest levels of AP and recall, while maintaining a high FPS. Further observation of Fig. 10 clearly reveals varying degrees of FPs and missed detections in the results of other algorithms. In summary, the proposed algorithm demonstrates high accuracy and robust generalization capability for detecting small ship targets in low-resolution remote sensing images. However, in this dataset, the presence of a few remote sensing images containing relatively large ship targets directly impacts the performance evaluation of the model. Specifically, large ship targets occupy more pixel space in the images, which enables the model to capture their features more easily. Consequently, the AP value reaches 96.12%, surpassing that of the DOTA-v2.0-Ship dataset. Larger targets enhance detection accuracy.

Furthermore, to validate the robustness of the algorithm, we conducted comparative experiments on the Ship-Own dataset with five outstanding algorithms: COCO-Net LMO-YOLO, DiffusionDet, HFPNet, YOLO-SM, and ours. The detection performance is shown in Table IV. Meanwhile, the LRS^2-DM algorithm was also visualized on the Ship-Own dataset, as shown in Fig. 11.

A comparative analysis of different algorithms clearly shows that LRS^2-DM consistently achieves the highest mAP and AP(s) values across all tests. This result demonstrates the exceptional performance of the LRS^2-DM algorithm in ship

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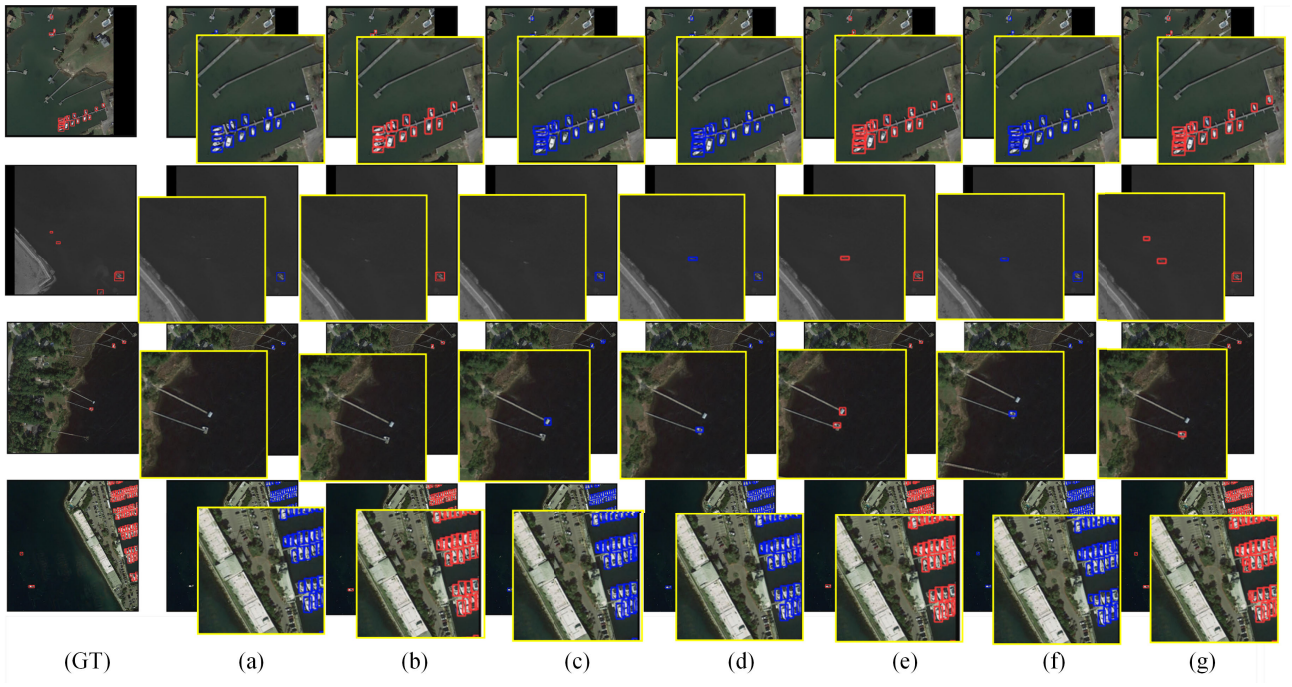


Fig. 9. Visualization of various algorithms on the DOTA-v2.0-Ship dataset. (a)–(g) Visualized detection results of DCNet, DiffusionDet, COCO-Net, LMO-YOLO, HFPNet, YOLO-SM, and our method, respectively.

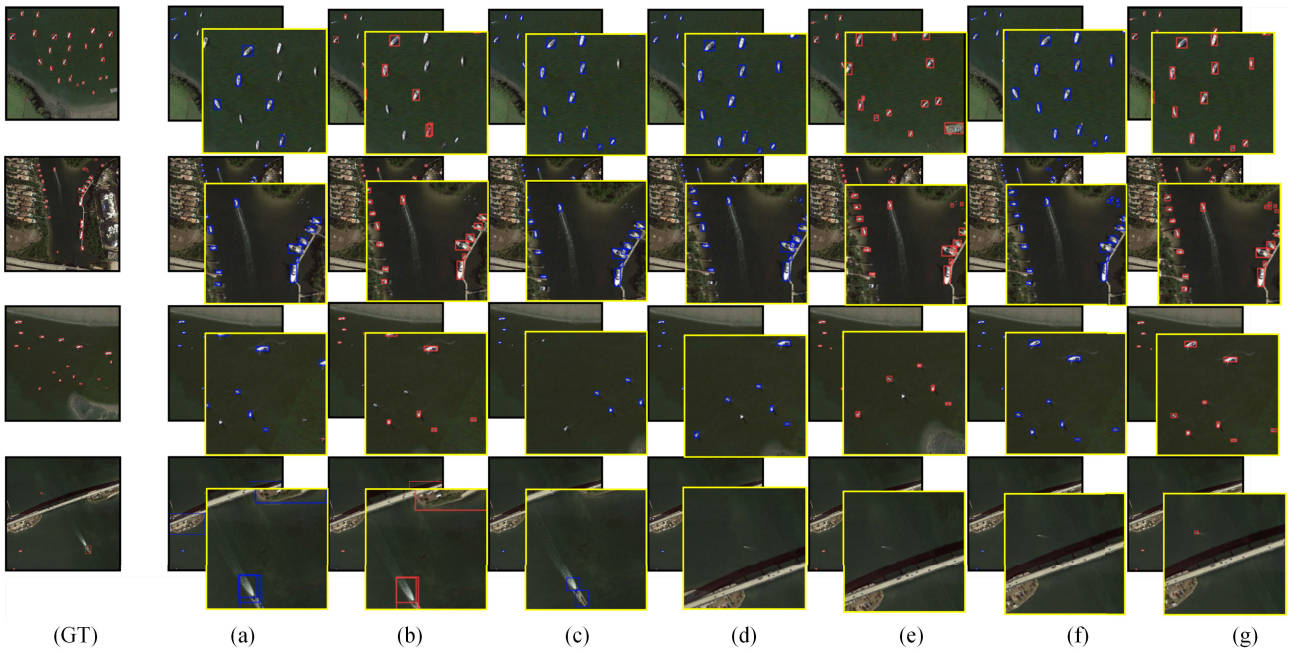


Fig. 10. Visualization of various algorithms on the S-Ship dataset. (a)–(g) Visualized detection results of DCNet, DiffusionDet, COCO-Net, LMO-YOLO, HFPNet, YOLO-SM, and our method, respectively.

detection tasks, particularly in small-target detection, where it significantly improves detection accuracy. It indicates that the algorithm possesses strong robustness and efficiency, making it well-suited for practical applications in small ship detection scenarios.

E. Ablation Study

To address the challenges of detecting low-resolution and small-sized ship targets, this article proposes incorporating cognitive conditions as input information. By sequentially processing through an L^{2}SR module and an SRM, the detection accuracy of remote sensing ships is significantly improved. To precisely quantify the contributions of each module in the proposed method, we conducted ablation experiments on the three datasets, ensuring consistent training and data augmentation schemes. The experimental results are summarized in Table V.

On the three subdatasets, the baseline model yielded AP scores of 85.87% (AP_1), 86.15% (AP_2), and 83.46% (AP_3), along with an inference speed of 48.4 FPS. These results serve as a reference benchmark for evaluating the effectiveness of subsequent module configurations.

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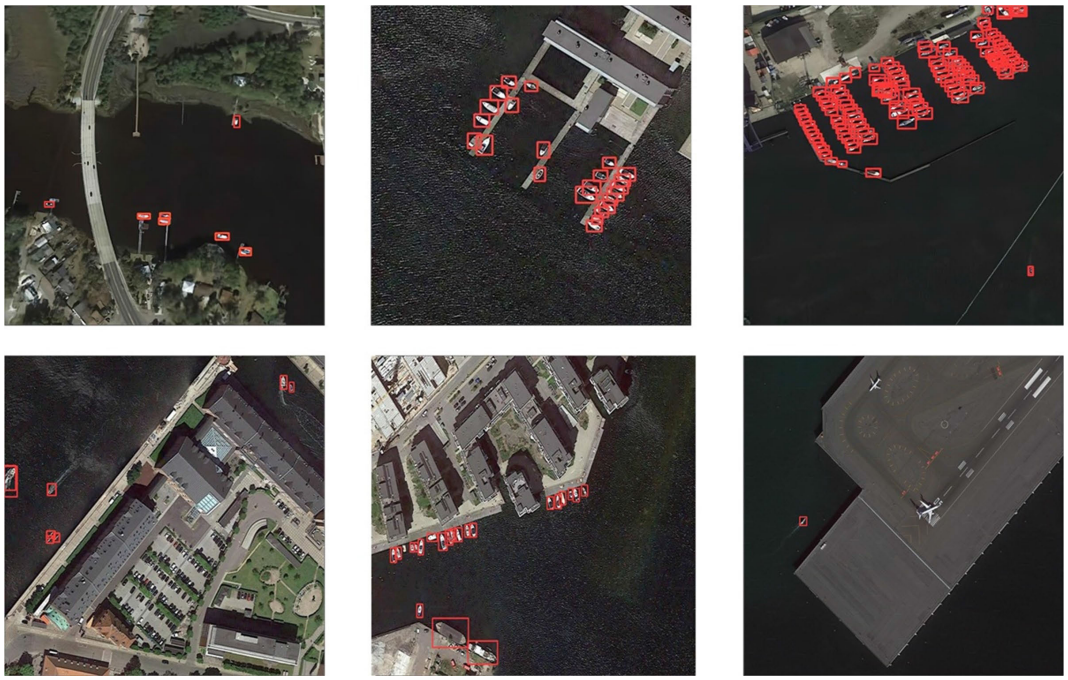


Fig. 11. Detection results of LRS^2-DM on the Ship-Own dataset. Detection results of LRS^2-DM on the Ship-Own dataset, showcasing the model's performance in identifying small-target ships in low-resolution remote sensing images.

TABLE V

ABLATION EXPERIMENTS ON THE DOTA-V2.0-SHIP, S-SHIP, AND SHIP-OWN DATASETS. AP_1, AP_2, AND AP_3 REPRESENT THE AP VALUES ON THE DOTA-V2.0-SHIP, S-SHIP, AND SHIP-OWN DATASETS, RESPECTIVELY. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Baseline	CCI	L^{2}SR	SRM	AP_1	AP_2	AP_3	FPS
✓				85.87	86.15	83.46	48.4
✓	✓			86.93	87.46	85.02	**55.2**
✓		✓		87.21	88.32	86.25	46.3
✓			✓	88.05	89.10	86.72	45.8
✓	✓	✓		89.10	90.18	87.54	<u>50.2</u>
✓	✓		✓	92.02	92.85	90.67	47.5
✓		✓	✓	<u>94.47</u>	<u>94.86</u>	<u>90.74</u>	44.6
✓	✓	✓	✓	**95.34**	**96.12**	**92.14**	49.2

With the introduction of the CCI module, the detection accuracy on all three datasets improved, with AP_1, AP_2, and AP_3 increasing to 86.93%, 87.46%, and 85.02%, respectively. Additionally, the FPS increased to 55.2, indicating that the CCI module not only provides semantic priors but also effectively reduces the computational burden during the inference stage. When the L^{2}SR module was introduced alone, detection accuracy further improved, with AP values of 87.21%, 88.32%, and 86.25% for the three subsets. However, the inference speed decreased to 46.3 FPS, suggesting that while the enhancement of image clarity improved object separability, it also introduced additional computational costs. After the introduction of the SRM module, all three AP metrics increased to varying degrees, demonstrating the effectiveness of this module in modeling fine details of small objects.

When both the CCI and L^{2}SR modules are introduced simultaneously, the model's AP values on the three datasets increase to 89.10%, 90.18%, and 87.54%, respectively, while FPS also improves to 50.2. Compared to using only the L^{2}SR module, this combination improves inference speed while maintaining accuracy gains, suggesting that the CCI module can partially compensate for the computational burden introduced by the image enhancement module. Further combining the L^{2}SR and SRM modules, detection performance continues to improve, with AP_1, AP_2, and AP_3 reaching 94.47%, 94.86%, and 90.74%, respectively. When all three modules are integrated, the model's AP values on the three datasets reach 95.34%, 96.12%, and 92.14%, the highest among all combinations, while FPS remains at 49.2, demonstrating excellent detection efficiency. This result indicates that the CCI, L^{2}SR, and SRM modules complement each other well in functionality, achieving the most balanced performance in terms of both accuracy and efficiency.

Furthermore, to visually illustrate the performance enhancement brought by the L^{2}SR module, we conducted comparative experiments using images reconstructed by this module and by other leading super-resolution algorithms, including MSAGAN [58], T-GAN [59], and DMSC-GAN [60]. Quantitative results of image enhancement are summarized in Table VI, while qualitative comparisons are presented in Fig. 12. As demonstrated in Fig. 12, images processed by the L^{2}SR algorithm exhibit notably superior clarity and spatial resolution compared to those produced by the other methods. Further analysis of Table VI supports this observation: L^{2}SR achieves the highest scores in both SSIM and PSNR.

Similarly, to validate the effectiveness of the SRM, we conducted feature visualization on the images processed by this module, as shown in Fig. 13. A comparison reveals that the small ship target features in remote sensing images processed by the SRM are more concentrated and distinct, with significantly enhanced and detailed features. This validates the effectiveness and practicality of the SRM. Moreover, it provides richer and higher quality feature information, improving the detection performance of small targets and ensuring more accurate and reliable results.

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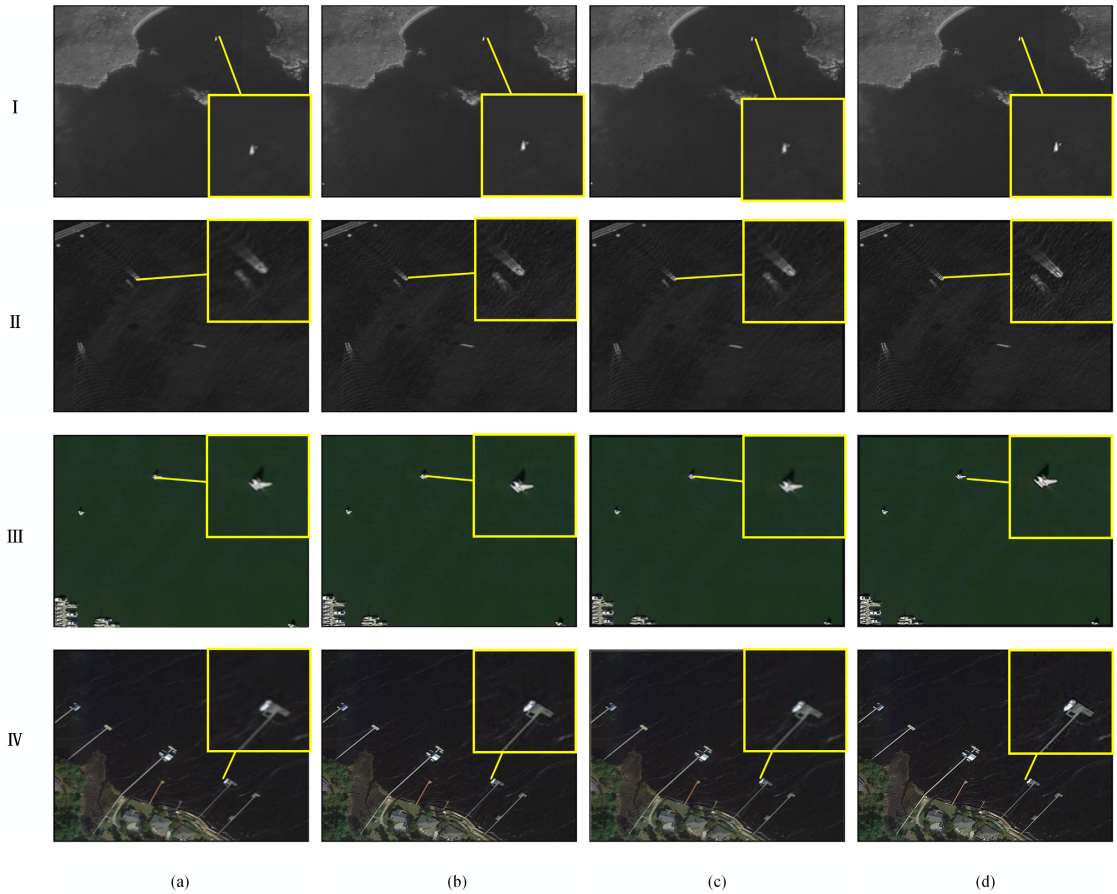


Fig. 12. Image enhancement example using the L^{2}SR module. (a)–(d) Correspond to the MSAGAN, T-GAN, DMSC-GAN, and L^{2}SR methods, respectively.

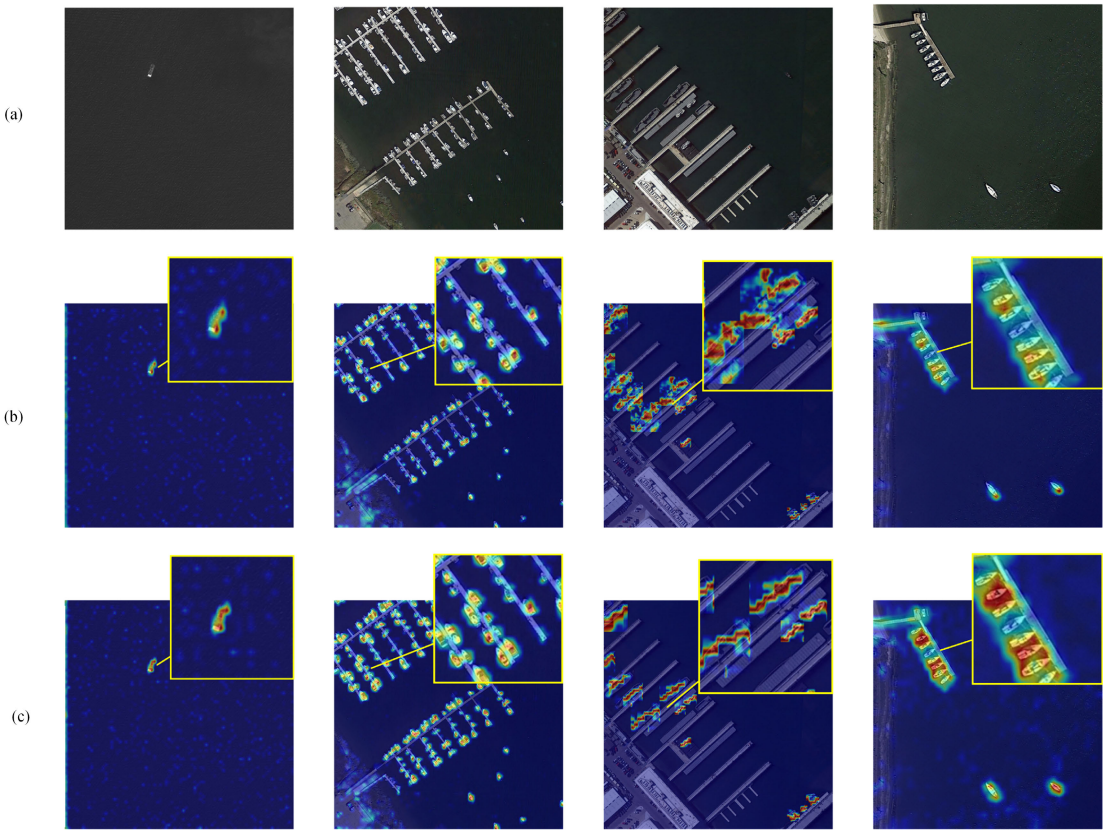


Fig. 13. Feature visualization of the SRM: (a) original image; (b) feature map of the original image; and (c) feature map after the SRM processing.

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TABLE VI

COMPARISON OF IMAGE ENHANCEMENT PARAMETERS USING THE L^{2}SR MODULE. THE MSAGAN ALGORITHM PERFORMS SECOND BEST IN TERMS OF SSIM, WHILE THE DMSC-GAN ALGORITHM ACHIEVES THE SECOND-BEST PSNR. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Image	MSAGAN		T-GAN		DMSC-GAN		L^{2}SR	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
I	0.9415	34.26	0.9345	34.25	0.9328	34.18	0.9528	34.92
II	0.9469	33.56	0.9416	33.69	0.9331	33.91	0.9502	35.03
III	0.9324	42.42	0.9286	41.96	0.9365	39.25	0.9454	35.13
IV	0.9298	33.78	0.9331	33.44	0.9267	32.94	0.9426	34.88
Average	<u>0.9377</u>	33.51	0.9345	33.88	0.9319	<u>33.95</u>	**0.9447**	**35.01**

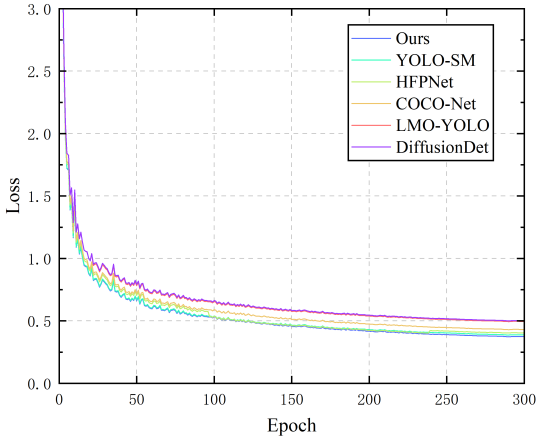


Fig. 14. Comparative performance of loss functions. A comparative loss analysis of YOLO-SM, HFPNet, COCO-Net, LMO-YOLO, DiffusionDet, and our method at epoch 300.

To evaluate the effectiveness of the proposed improvements to the loss function, comparative experiments were conducted with three widely used loss functions. As illustrated in Fig. 14, the proposed loss function exhibits superior convergence performance. In contrast, the other loss functions demonstrate slower convergence rates and lower convergence quality. These findings highlight the significant optimization advantages of the proposed method.

V. CONCLUSION

LRS^2-DM is a small ship target detection algorithm specifically designed for low-resolution remote sensing images. It integrates cognitive condition inputs, an L^{2}SR module, and an SRM to address challenges posed by low image resolution and the small size of ship targets. LRS^2-DM is a small ship target detection algorithm specifically designed for low-resolution remote sensing images. It integrates cognitive condition inputs, an L^{2}SR module, and an SRM to address challenges posed by low image resolution and the small size of ship targets. Subsequently, an L^{2}SR module is employed to enhance image clarity, effectively mitigating the loss of detail caused by low resolution. Furthermore, the LRS^2-DM algorithm introduces an SRM dedicated to improving edge texture detailed features of small ship targets, resulting in more precise and nuanced detection outcomes. This step is critical for enhancing the accuracy of small-target detection as it captures subtle differences in the shapes and structures of ships. To further enhance the model's stability and performance, the LRS^2-DM algorithm employs an improved loss function. This function is more rationally designed, effectively guiding the model to converge to better solutions during training. Consequently, it demonstrates superior robustness and generalization in practical applications. The experimental results demonstrate that the LRS^2-DM algorithm excels in addressing the challenges of low-resolution remote sensing images and small ship targets. Compared to other state-of-the-art detection algorithms, it exhibits significant advantages, achieving superior ship detection performance and providing novel approaches and methodologies for small-target detection in remote sensing images.

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